

# CSE482 Class Project Sports Analytics

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## Problem definition:

- Data Collection: Gather sports-related data from reliable sources such as official sports platforms, APIs, and public datasets, focusing on one baseball/soccer.
- Performance Metrics Definition: establish relevant sport-specific metrics
  - How often the player gets a hit
  - How often the player gets on base
  - How much power the player has in hitting
  - Combines ability to get on base and power
  - How patient a hitter is (draws walks)
  - How often the player strikes out
  - How many runs the player helps create
  - Overall hitting impact (all hits weighted)
- Statistical and Predictive Modeling: Use algorithms and statistical techniques to build models that forecast match outcomes, player performance trajectories, or team standings.
- Insights Generation: Provide recommendations and interpretations based on findings for practical use by coaches, analysts, or sports enthusiasts.

## Where you get the data

Our datasets originated from <https://www.rebas.tw/> which is a startup team in Taiwan focused on developing sports data technology and building a baseball fan community.

| Stat | Description                   |
|------|-------------------------------|
| PA   | Number of times a batter bats |

|          |                                                 |
|----------|-------------------------------------------------|
| AVG      | How often a batter gets hits                    |
| OBP      | How often a batter gets on base                 |
| SLG      | How much power a batter shows with hits         |
| ISO      | Extra-base hit power (excluding singles)        |
| BABIP    | Hits when ball is put in play (not HR)          |
| BIP%     | Percent of balls put in play                    |
| OPS      | Overall hitting skill (OBP + SLG)               |
| OPS+     | OPS compared to league average (100 = average)  |
| tOPS+    | OPS in specific situation versus normal         |
| RC       | Estimated runs contributed by the batter        |
| wOBA     | Weighted on-base (values extra-base hits more)  |
| BB%      | How often walks happen                          |
| BB/K     | Walks compared to strikeouts                    |
| K%       | How often strikes out happen                    |
| PutAway% | How often 2-strike pitches result in strikeouts |